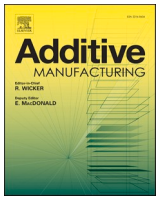
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Methodology to determine printability criteria of highly concentrated pastes through rheological characterization

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ABSTRACT

Material extrusion is an additive manufacturing technique that enables the creation of reproducible and complex hardware by depositing a viscous, shear-thinning ink onto a substrate in a custom-pattern via extrusion through a syringe. The ability of an ink to be extruded onto a substrate in many layers, and maintain the desired shape is what defines the printability. Printability is often investigated by formulating, printing, and postmortem analysis of final parts in an iterative manner. Investigations of printability through rheological characterization have often been concerned with inks that straddle the line between printable and too thin, leaving out an entire class of inks that are highly-filled pastes, where extrudability is the limiting factor. Highly-filled pastes continue to pose issues for researchers as the effect of filler morphology, size, loading, and packing fraction on the ink rheology and corresponding printability is not understood. While traditional rheological characterizations may be useful for some inks, we show that protocols utilizing steady-shear, or large-amplitude oscillatory shear are difficult and unreliable for highly-filled pastes. Through transient rheology paired with real-time images we show that each traditional protocol produces inhomogeneous deformations that violate the assumptions that underly common rheological definitions. Instead, we demonstrate metrics measured with small-amplitude oscillatory shear that are correlated to the printability of various ink formulations ranging in loading. The rheological measures that accurately predict the printability of the inks are the axial stress measured at small amplitudes, and the critical stress amplitude above which rheological characterizations become impossible. In addition, we estimate the maximum packing fraction for each filler, based on the exponent common to hard sphere models, and show that the printability of each ink can be predicted by the ratio of the packing fraction to the theoretical maximum. We show how small-amplitude oscillatory shear allows users to develop printability criteria for any ink to enhance the workflow in the development of new inks, increase the performance of material extrusion printing, and improve the stability of printed parts, with less wasted time and materials.

1. Introduction

There has been increasing interest in additive manufacturing (AM) due to the ability to utilize unique inks and produce custom-designed parts in complex shapes that are impossible to attain with traditional manufacturing. In the case of direct-ink write (DIW) material extrusion AM, an ink is extruded through a nozzle that is attached to a moveable head to generate printed parts in a layer-by-layer fashion. Most DIW inks are suspensions of particles - such as ceramics [1,2], metals [3,4], and organics [5,6] – in a base – such as silicones [7], epoxies [8], or urethanes [9]. Due to the variety of material types, the applications are just as broad, ranging from building materials [10–13], geology [14–16], and energy storage [17,18] to medical devices [19–21] and biology [5,6]. While each application may have unique requirements, an understanding the ability of an ink to be extruded and hold its shape, or the printability, is crucial to the development of new inks and the stability of 3D-printed parts. Printability has remained a poorly defined term, but it is generally accepted that an ink is printable if: it can be extruded through the nozzle in a timely manner, can stick to a build plate, and produce a part that holds its shape after extrusion has concluded. The ability to extrude an ink is often represented on a rheometer by the inks response to the application of a steady shear rate, while the ability to

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sustain the weight of subsequent layers is often related to a yield stress or shear modulus. Thus, there are two ways that an ink becomes non-'printable'. Either an ink is 'runny' and cannot support the subsequent weight of additional layers, or an ink is 'jammed' and cannot be extruded at all. Definitions of these terms for this work are included in Section 3.1. There are many works that look at printability for inks that straddle the line between 'runny' and 'printable [12,22–25]. There remains, however, another end of the spectrum of printability for inks at the higher end of loading, between 'printable' and 'jammed.' Thus far, there has been no discussion about printability criterion for these types of inks. These highly-filled materials are important for a range of applications, especially for inks with functional particles. For example, in batteries, the volume fraction of active particles determines the capacity of a printed electrode [26,27]. In the field of medical devices, compressive and flexural strength of dental composites is directly proportional to the filler loading [28,29]. Concrete strength can also be increased through filler loading [30,31]. There are even discussions of space-based 3D printing using fillers such a moondust [32,33] where the liquid composition needs to be minimized. Some applications currently utilize a 'densification process' where binder or porosity is removed after the printing has completed [34–36]. If particle loading could be maximized prior to printing, this would reduce the cost of this type of post-processing.

Rheology is a tool often used to characterize DIW inks, as it is well suited for studying the behavior of complex fluids. Much research has been done to try and determine generalizable printability criteria through different rheological tests. Since many of these works are concerned with determining printability in inks that are on the border between 'runny' and 'printable', printability is often defined as a minimum value of viscosity [12,22,23], modulus [17,24,37], yield stress [12,17,23,25,37,38], or some other measure related to energy dissipation [12,23,24], above which an ink will hold its shape and below which it will not. If criteria for printability can be determined, users can determine *a priori* if an ink formulation will be printable based on simple rheological characterizations rather than printing numerous formulations on a printer, thus saving time, money, and resources.

There are two main rheological protocols that have been utilized in attempt to define printability criteria: shear rate sweeps, and oscillatory amplitude sweeps under stress or strain control. In an oscillatory amplitude sweep, a stress- or strain-controlled oscillation is applied to the ink at a set frequency and amplitude until steady alternance (i.e. steady-state) is achieved. Then, the amplitude of the oscillation is increased stepwise, commonly spanning several orders of magnitude. For each amplitude, period-average metrics, such as the storage and loss moduli (G' and G''), are reported once the rheological software has determined steady alternance has been reached, or a user-specified amount of time has passed. If the amplitude of oscillation is small, the rheometer oscillates about the equilibrium position of the ink and the equilibrium structure can be probed. This is said to be the linear viscoelastic response of the material. From the linear viscoelastic response one can define a storage and loss modulus, which are related to the energy stored and dissipated by a material under a prescribed deformation. The storage modulus is a commonly used rheological variable for determining the printability of an ink, however reported 'printability' values span an order of magnitude, rendering this metric insensitive. [17,24,37].

The other protocol often used to characterize DIW inks is a shear rate sweep, where a constant shear rate is applied, and the average stress is reported after the rheological software determines steady-state has been achieved, or after a user-specified amount of time has passed. The shear rate is then step-wise ramped up or down, and the average stress at each shear rate is determined. This stress is then used to calculate an apparent viscosity. If the shear rate is applied long enough for a material to acquire 3–5 strain units, the steady-state response of the material can be determined. Plotting the steady-state stress response for a range of applied shear rates results in a flow curve. Typically, a yield stress is determined as the low-rate limit of the steady-state stress, or the fitting of a power-law equation, such as Bingham or Herschel-Bulkley [39,40].

There is no consensus on the best or most appropriate method of determining a yield stress [41–47]. This is best exemplified by the work of Dinkgreve et al. [47], where the authors show many yield stress values determined for a simple yield stress fluid across different protocols span half an order of magnitude between steady-shear, oscillatory, and creep experiments. For DIW inks, the reported yield stresses that make an ink printable as determined by steady-shear range between 200 Pa to 2000 Pa [12,38,48], and 200 Pa to 500 Pa [17,24,37] for oscillatory shear. This leads to metrics of printability that range by a factor of five, while the method to determine this parameter depends heavily on the protocol and data interpretation. This has led to many results for printability being highly specific to the inks tested, without generalizability to other systems. While other works on DIW ink printability look to be more general and define inks to be printable if the yield stress is between 0.5 and 1 times the density times the final part height times gravity [23,25,37,75,76]. However, this criterion makes no consideration of the inks themselves which are highly-filled, where extrudability is the main factor determining printability.

In this article, we investigate the properties of three DIW inks, each a suspension of hard particles in a binder. Each ink formulation has been categorized as 'runny', 'printable', or 'jammed', through extrusion and measurement of slumping. We pair rheology with real-time imaging of the sample to demonstrate why steady-shear and oscillatory amplitude sweeps are not suitable for highly-filled inks. We look at the response of the DIW inks to small-amplitude oscillatory shear and discuss the measures that can predict the printability of each ink, and that do not. We determine the critical stress and strain beyond which the rheological characterizations are difficult, and why that is the case. Finally, we utilize a modified reduced modulus to define a new printability criterion that collapses data for inks made with three different particle fillers with complex differences in size distribution and morphology. This methodology can be utilized to determine printability criterion for any print condition for hard-particle DIW inks.

2. Materials and methods

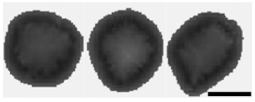
2.1. Ink formulation and particle characterization

Direct-Ink Write formulations were created using the silicone base (Base, Sylgard™182, Dow Chemical $\rho = 1.03$ g/cm^3 [49]), and three different filler particles having different morphologies and size distributions. Inks filled with calcium carbonate (Acros Organics) were prepared ranging from 46 vol% to 54 vol% (69 wt% to 77 wt%). Inks formulated with melamine (Acros Organics) had volume fractions from 56 vol% to 74 vol% (71 wt% to 81 wt%). Finally, powdered aluminum (Valimet) inks were formulated ranging from 40 vol% to 54 vol% (63 wt % to 75 wt%) filler.

The various filler materials were analyzed for size distribution and morphology using a Morphologi 4-ID (Malvern Panlytical) fixed with a dispersion unit cell. A consistent volume of 1 mm^3 was maintained for each particle analyzed. For 30 s, a pressure of 0.8 bar of nitrogen gas was applied to disperse the particles onto a 2 in. × 4 in. quartz plate, followed by 3 min of settling time before analysis was conducted at a 50× magnification. Densities, particle sizes, aspect ratios, and example images of each particle are shown in Table 1. The calcium carbonate filler particles are irregularly shape and jagged, while melamine particles are rounded and oblong, and the aluminum particles are round spheroids. The mean and standard deviation of the particle size and aspect ratio are shown in Table 1 while the full distributions are shown in the Supplemental Material (SM), Figure SM1.

Each ink was prepared in a planetary mixer (THINKY ARE-310), first by adding the silicone base, then half the required filler. This sample was mixed at 2000 RPM for 2 min. Then, the remaining filler was added, along with an yttria-stabilized zirconia ball bearing of 1 cm in diameter.

Table 1
Density, mean and standard deviation of particle size and aspect ratio, along with exemplary images of each filler particle. Particle size, aspect ratios, and images were obtained on Morphologi 4-ID. Scale bars in each image set represent 1 micron.

Material	Density (g/cm^3) [50–52]	Circle-Equivalent Particle Diameter		Aspect Ratio		Particle Examples
		Mean (μm)	Standard Deviation (μm)	Mean (-)	Standard Deviation (-)	
Calcium Carbonate	2.93	1.71	1.47	1.24	0.136	
Melamine	1.57	6.46	4.28	1.29	0.143	
Aluminum H2	2.7	2.09	2.31	1.20	0.141	

These samples were then mixed an additional eight minutes at 2000 RPM in two-minute intervals with hand mixing between each. While mixing led to an increased temperature, all inks were allowed to cool before any printing or rheological measurements. Immediately prior to rheological testing, samples were mixed for two minutes to achieve a homogenous consistency. No particle setting was observed on the timescale of the rheological experiments performed, nor the timescale of DIW printing.

2.2. Rheology and real-time imaging

Rheology experiments were performed on two stress-controlled rheometers, the MCR 702 (Modular Compact Rheometer 702, Anton Paar) and DHR3 (Discovery Hybrid Rheometer 3, TA Waters) with 40 mm profiled/cross-hatched parallel plates. Prior to any experiment, the rheometer and test fixtures were calibrated for inertia and friction and the gap was zeroed. For all rheometer samples, the gap was lowered to 1.05 mm, ink was trimmed, and the gap was reduced to the target value of 1 mm. Following the reduction of the gap, all samples were held at zero shear stress for 5 min. Other geometries were considered, such as a cup-and-vane or standard Couette cell. Ultimately cross hatched parallel plates were chosen because they allowed for clear visual cues as to when flow inhomogeneities occurred, which is difficult in cup-and-vane. Additionally, the cross hatched plates helped avoid wall slip, which was seen in the Couette cell.

On the MCR 702, steady-shear experiments were performed at 0.1 s^{-1}, and oscillatory experiments were performed at a constant stress amplitude of 63.1 Pa and frequency of 0.1 Hz, both with a 1 mm gap. The rheology was coupled with real-time images of the inks under the applied conditions. The images were captured with a FLIR Blackfly camera (S BFS-U3–123S6C). Rheology measured on the MCR, and the corresponding images are shown in Fig. 1 and Fig. 2.

The experiments performed on the DHR3 were stress-controlled oscillatory experiments performed at a frequency 1 Hz and tested over a two-decade range of stress amplitudes chosen to bracket the yielding transition for each ink. The strain amplitude was measured for each period of oscillatory stress and tracked transiently over the course of each stress amplitude. All reported values are calculated from the TRIOS software (TA Waters), and the data are shown in Fig. 3 through Fig. 8.

3. Discussion

3.1. Definition of printability and ink extrusion shear rate

To determine criteria for printability, we must rigorously define it.

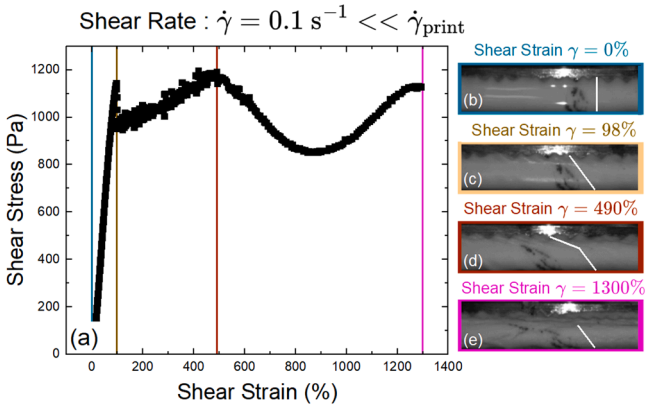


Fig. 1. Steady-shear startup at $\dot{\gamma} = 0.1\ s^{-1}$ for a 'printable' 52 vol% $CaCO_3$ ink. (a) Transient stress measurements, (b)-(e) Images corresponding to various strain values indicated by vertical lines in (a). White guide lines are shown along with the images to guide the eye.

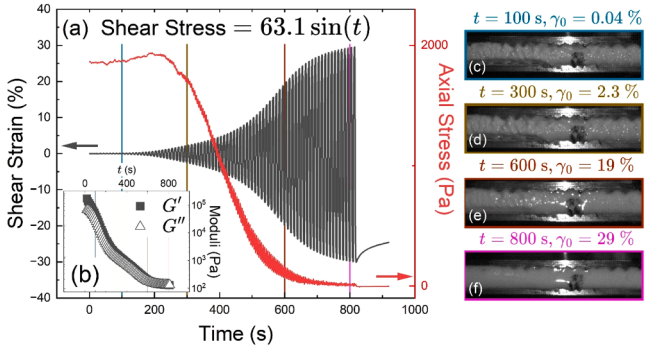


Fig. 2. Stress-controlled oscillatory experiment performed at 0.1 Hz and 63.1 Pa for a 'printable' 52 vol% $CaCO_3$ ink. (a) shows the transient strain response corresponding to the left axis, and the axial stress, on the right axis. Inset (b) shows the storage and loss moduli (G' and G'') over this same time. (c)-(f) show real-time images of the ink surface corresponding the time points denoted as vertical lines in (a).

For this work, we are defining an ink to be printable if it satisfies two criteria: 1) the ink must be able to be extruded out of a syringe with a modest nozzle diameter and flow rate (0.84 mm nozzle at 0.25 mL/min) and a reasonable load cell (50 kg), and 2) the height of a printed cone

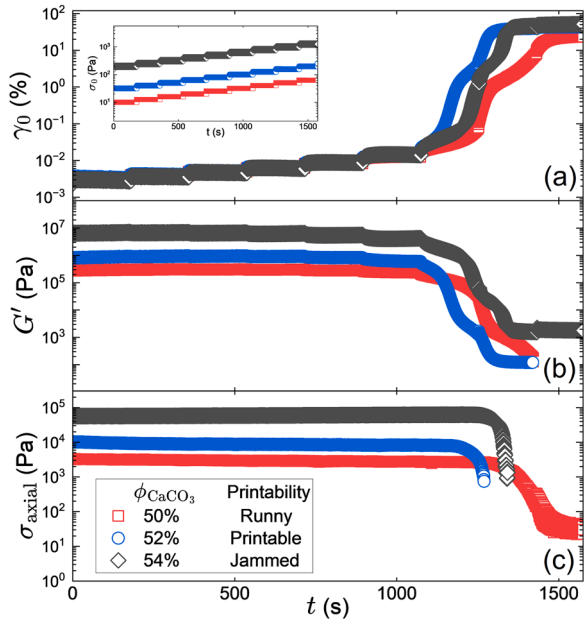


Fig. 3. (a) Measured strain amplitude (γ_0), (b) storage modulus (G'), and (c) axial stress (σ_{axial}) for three calcium carbonate inks at 50 vol% (red squares), 52 vol% (blue circles) and 54 vol% (black diamonds), which are 'runny', 'printable' an, 'jammed' respectively, under the stress amplitudes shown in the inset of panel (a). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

must change less than 10 % one minute after printing has been concluded, as determined by a change in the height of the extruded ink. If an ink cannot satisfy criterion 1) we say it is 'jammed'. If an ink cannot satisfy criterion 2) we say it is 'runny'. If an ink can satisfy both criteria, the ink is 'printable'.

The extrudability portion of this also defines a range of shear rates an ink will experience while being printed. The apparent shear rate at the wall for a syringe nozzle can be calculated as [53],

$$\dot{\gamma}_{aw} = \frac{4Q}{\pi R^3}, \tag{1}$$

where $\dot{\gamma}_{aw}$ is the apparent shear rate at the wall, Q is the volumetric flow rate, and R is the radius of the nozzle tip. There are a range of possible correction factors based on rheology and slip, but a range for the shear rate experienced by the pastes at the syringe tip would be between 10 s^{-1} and 100 s^{-1}, that can be directly applied on a rheometer through a steady-shear experiment [37].

3.2. Evaluating common protocols utilized to predict printability

Shear rate sweeps are often utilized to determine the yield stress of DIW inks by plotting the steady-state stress versus the shear rate. A typical steady-shear flow curve is constructed by choosing a range of strain-rates to test and subsequently measuring the corresponding stress once steady-state is achieved. This implies that the rheometer will correctly identify steady-state, and that the resulting deformation field is homogenous. On a flow curve, a single stress value is reported for an applied shear rate. However, this implies the rheometer or user has correctly identified steady-state conditions, at which point the rheometer reports a single stress value per the applied shear rate.

Fig. 1 demonstrates why allowing the rheometer to detect equilibrium is unsuitable for highly-filled inks. Fig. 1(a) shows the transient stress response of a 'printable' 52 vol% calcium carbonate suspension, at a shear rate of 0.1 s^{-1} over the course of 130 s. Real-time images of the ink under this strain rate are shown in Fig. 1(b)-(e). The edge of the sample was marked with a black line to aid the image analysis, and white lines have been added to the figure to guide the reader. Fig. 1(b) shows the ink at rest, prior to the application of a constant shear rate, at which point the top plate begins to rotate clockwise, or to the left in the image, while the bottom plate remains stationary. This is indicated visually by the vertical white line in the image. In Fig. 1(a), from 0 % to 98 % strain the ink deforms elastically, as shown by the constant slope of the stress versus strain in this region, and Fig. 1(c) also shows lines of constant slope across the material. Immediately after reaching 98 % strain, the stress response drops rapidly, then begins to increase again at a different constant slope. The reason for this behavior can be seen in Fig. 1(d). The top half of the sample has continued to response elastically, as indicated by a steeper white line, however, the bottom half has remained in a similar state to that seen in Fig. 1(c). This change in sample behavior is indicative in inhomogeneous behavior, that could only be understood through transient tests and real-time imaging. This second response continues until 490 % strain is reached, where the stress then begins to behave non-monotonically. The reason for this behavior in the stress can be seen in Fig. 1(e), at 1300 % strain, where the ink has fully separated into two regions, as a large slip plane has formed directly beneath the moving top plate. For a steady-shear test to be meaningful, the flow field must be homogenous, and here we clearly show that is not the case. Thus, we conclude that the shear rate is not homogeneous across the gap. No further rheological characterization could be performed on the ink as much of the material had left the gap.

The applied shear rate in this experiment was 0.1 s^{-1}, which is 100× smaller than the shear rates experienced by an ink in the nozzle while printing. If this were programed as a shear rate sweep and the user was not tracking the transient stress response or visually examining the ink, the rheometer would move on to the next shear rate(s) and report a single value of stress for each corresponding shear rate in the procedure. This figure illustrates why performing a flow sweep on the rheometer without looking at the time-dependent response of the stress is insufficient and can lead to inaccurate conclusions. Additionally, steady-shear is not suitable for highly-filled inks because slip planes and fracturing occur before any steady flow can be achieved. Therefore, a yield stress for these highly-filled inks could not be determined from a steady-shear sweep.

While steady-shear experiments apply a constant shear rate, and measure the resulting stress, one can just as easily apply a constant stress, and measure the evolution of the strain and shear rate. The application of constant stress is a creep test, and has also been used to determine yield stresses in the past [47]. In a creep test, the evolution of the strain and strain rate are measured over time. Below the yield stress, the strain rate continually decreases toward zero. By iteratively increasing the stress, the response of the material is seen through changes in the shear rate from decreasing to constant, at which point the yield stress can be determined [47,54,55].For these highly-filled pastes, creep tests eventually yield the same response as the steady-shear. At some point, the material will be unable to sustain the stresses being applied, and rather than homogeneous rheometric flow, the material begins to leave the gap. In the case of a creep test, the shear rate will increase exponentially as the rheometer tries to apply a constant stress to an ever-dwindling amount of material, eventually leading to an instrument overspeed. An example of this is seen in Figure SM 2. While this work does not investigate the creep response of highly-filled inks, creep tests, these tests may offer another way to measure the transient behavior of these inks. As creep tests are often paired with a recovery step, the new field of recovery rheology pairs extremely well with these protocols, and has recently been utilized to examine the transient yielding of many different yield stress fluids, opening the door for further investigation [56–59].

Another common protocol to determine a yield stress is through an oscillatory shear amplitude sweep. For each amplitude a constant stress or strain is applied, and from the measured corresponding strain or stress, the software will report the amplitude of the response, the storage and loss moduli, and other metrics, all of which are measures of the

average periodic response. Fig. 1 shows what issues can arise if the transient behavior is not examined. Similar issues can also arise during oscillatory testing. At small stresses or strains within the linear viscoelastic plateau, the ink is well-behaved and reaches steady-state within ten to twenty periods. However, once the applied stress or strain becomes large enough, the ink response is difficult to measure accurately.

Fig. 2 shows the response of a 'printable' 52 vol% calcium carbonate paste to a constant stress amplitude of 63.1 Pa over 80 periods at 0.1 Hz, along with real-time images of the material under application of this oscillatory stress. Included in these images is a vertical black line added to the material, to allow tracking of the flow field. Fig. 2(a) shows the transient strain response on the left axis, as well as the axial stress on the right axis. Fig. 2(b) shows the storage and loss modulus for each period, over the entire 800 s experiment. Fig. 2(c)-(f) shows real-time images of the material at different points in time. Fig. 2(c) shows an image of the ink after 100 s ($\gamma_0 = 0.04$ %, $G' = 44829$ Pa, $G'' = 30577$ Pa), where there are ridges in the ink, which were acquired during the trimming of the sample, and the material looks stable in the image. As the ink undergoes further oscillations, the strain amplitude begins to increase. Fig. 2(d) shows the ink after 300 s, the strain amplitude increases two orders of magnitude ($\gamma_0 = 2.3$ %, $G' = 1711$ Pa, $G'' = 1348$ Pa) but the material looks physically similar at the surface. By 600 s, the strain amplitude increases another two orders of magnitude ($\gamma_0 = 19$ % , $G' = 197$ Pa, $G'' = 180$ Pa) and in Fig. 2(e) the ridges begin to disappear and the surface appears glossy. At this point, during the oscillations, one can visually see that the flow is inhomogenous. The portion of the sample closer to the plate oscillates as expected, however the bottom half of the sample remains stationary. Finally, after 800 s the strain amplitude begins to level off ($\gamma_0 = 29$ %, $G' = 129$ Pa, $G'' = 131$ Pa) and in Fig. 2 (f) all the surface ridges are gone, and the ink surface is smooth, and the material has spilled over the bottom plate. Images and videos of the material can be seen in the SM, which show the full set of oscillations where this stress was applied.

If this were a traditional amplitude sweep, the rheological software would report a single value for strain amplitude, storage modulus, and loss modulus. However Fig. 2(a) and Fig. 2(b) show that there is much more to this picture. If the response were reported after 100 s, the moduli would appear to be on the order of 10^4 Pa, and since $G' > G''$, some would say it has not yielded. However, if the response were reported after 800 s, the moduli are more than two orders of magnitude smaller, around 10^2 Pa, and $G'' > G'$. While this may sound like the features of a material yielding, the real-time video shown in the SM of the material shows that there is not homogenous flow occurring. The paste in the lower half of the gap remains stationary, while the top plate oscillates. If the flow were homogeneous, we would expect the vertical black line to also show homogenous deformation. Fig. 2 shows images taken throughout the oscillations, and the time-resolved rheology. Fig. 2 shows oscillatory shear outside of the small-amplitude regime for highly-filled DIW inks is ineffective, because the applied oscillations produce non-homogeneous flow conditions which invalidate the assumptions which underly the common rheological calculations. Therefore, care should be taken to only test the small-amplitude response of highly-filled DIW inks such as those described here.

In addition to the strain response, the right axis of Fig. 2(a) shows the corresponding axial stress. The axial stress is calculated from the axial force measured by the rheometer, which is then normalized by the size of the plate being used. When paired with the images, Fig. 2 shows that a drop in the axial stress indicated that the material had started to leave the gap to escape internal stresses, which can be seen in Fig. 2(f) as the material in spilling over the edge. For this 52 vol% calcium carbonate ink, the stress amplitude of 63.1 Pa corresponds to the large-amplitude oscillatory shear region of rheological response. Therefore, large-amplitude oscillatory shear measurements need to be approached with great caution for highly-filled suspensions, especially if there is a large drop in the axial stress. In Section 3.4, we discuss this point more clearly when we define the critical stress.

3.3. Determination of a printability criterion using small-amplitude oscillatory shear

We have shown that both steady-shear and large-amplitude oscillatory shear rheology are ineffective for characterizing a highly-filled ink, even if that ink is otherwise 'printable.' With these techniques, the material fractures or changes and the resulting inhomogeneous flow means that rheological metrics are not well defined. Therefore, these protocols cannot be used to determine a yield stress or any other material property that could be correlated to printability. For the case of small-amplitude oscillatory shear prior to the drop in the axial stress, there is high confidence in the measured response of the rheometer representing the ink response. After the decline in the axial stress, corresponding to the ink leaving the gap, the values reported by the rheometer are not measures of the material properties, but a measure of a changing volume of material.

To correlate measures to printability, we performed stress-controlled oscillatory testing at a range of stress amplitudes each at 1 Hz, for 180 s, beginning with small amplitudes, and increasing. An example of the responses measured for ink formulations with three concentrations of calcium carbonate can be seen in Fig. 3. Fig. 3(a) shows the measured strain amplitude over time at nine different (increasing) stress amplitudes, each of which can be seen in the inset. For all three inks, which span the range of printability from 'runny' to 'jammed', the strain increases linearly as expected for the smallest stress amplitudes, followed by some critical stress amplitude that the material strain response begins to rise rapidly. Inks that were 'runny' slumped more than 10 % after printing, while inks that were 'jammed' exceeded 50 kgf when trying to print. 'Printable' inks were extrudable and slumped less than 10 %.

Fig. 3(b) shows the storage modulus evolving over time and Fig. 3(c) shows the axial stress. We see similar behavior for all three categories of printability: in all three measures, the small-amplitude response is constant and then there is a critical stress where the behavior changes and the axial stress drops during the application of this critical stress indicating that material is leaving the gap. These tests were all performed in triplicate, and the results with error bars are shown in Figure SM3. Results for the remaining calcium carbonate inks, as well as the melamine and aluminum inks are shown in Figures SM 4 – SM 8. We see a linear plateau response (such as the response seen in Fig. 3 at times less than 500 s), while the analysis of the large-amplitude oscillatory response (1000–1500 s) will be discussed in the next section.

The response of the inks to small-amplitude oscillatory shear represents the linear viscoelastic response, in which the equilibrium structure of the ink is being probed. The protocol described with Fig. 3 was applied to inks with three different types of filler and many levels of loading. Each was categorized as 'runny', 'printable', or 'jammed', by the criterion discussed in Section 3.1. For the metrics chosen, the linear viscoelastic data are presented in the form of vertical box plots, with each data point overlaid and is displayed in Fig. 4. The red box plots with squares correspond to inks that are 'runny', the blue box plots with circles correspond to inks that are 'printable', and the black box plots with triangles correspond to inks that are 'jammed'. The median of each category is a thick line across the box. The minimum and maximum of each data set is indicated by the whiskers of each box. Therefore, if a rheological measure distinguishes between all three printability responses (i.e., there is no overlap in the whiskers of each box), we determine it is a full predictor of printability and the panel has a transparent green filling. For metrics that were able to distinguish between inks at the higher end of loading (between 'printable' and 'jammed') we have indicated this with a gray fill, and diagonal lines.

As discussed in the introduction, the measure from small-amplitude oscillatory shear, which is most often correlated to printability, is the storage modulus. Other common measures for small-amplitude oscillatory shear are the loss modulus and the ratio of the loss modulus to

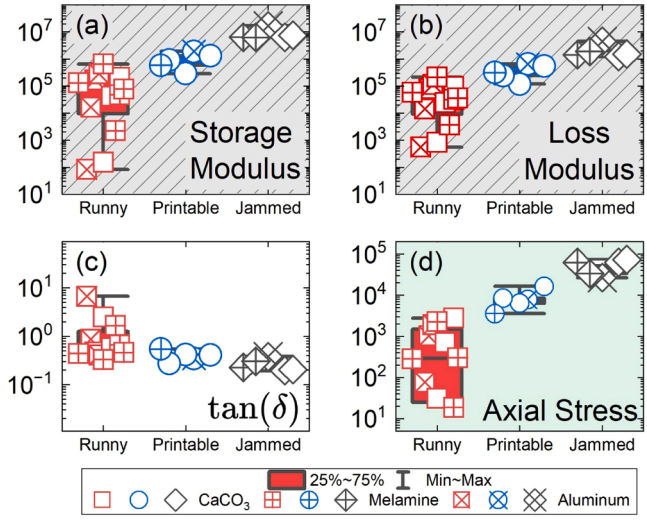


Fig. 4. Box plots showing the small-amplitude oscillatory shear responses of each ink at a frequency of 1 Hz, separated into the categories of 'runny', 'printable', or 'jammed'. The ink functions are (a) linear storage modulus in pascals (G'), (b) linear loss modulus in pascals (G''), (c) the ratio of loss modulus to storage modulus ($\tan(\delta)$) and (d) the axial stress in pascals (σ_{axial}). The axial stresses were taken when the applied stress amplitude was small, and the measured strain amplitude was less than 0.01 %. The boxes with diagonal lines represent a measure that was able to differentiate between inks that were 'jammed' and those that were 'printable' but was not able to determine between inks that were 'runny' and inks that were 'printable. The axial stress (green shaded background) is the only measure that was able to differentiate between all three ink types. Each filler type is differentiated by the symbol fill, 'open' represents calcium carbonate, '+' represents melamine, and 'X' represents aluminum filler. Eaching printability response is differentiated by symbol shape and color, 'runny' inks are shown by red squares, 'printable' inks by blue circles, and 'jammed' inks by black diamonds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

storage modulus, or $\tan(\delta)$. Fig. 4(a) and Fig. 4(b) show the linear viscoelastic storage modulus and loss modulus separated by printability. Between all three filler types, the storage and loss moduli successfully differentiated between inks that are 'jammed' and 'printable' but could not differentiate between inks that are 'runny' and 'printable'. This may explain the lack of consensus between the literature as to the required minimum value of the storage modulus for printability[17,24,37], since some inks in this work are 'printable' with a smaller storage modulus than inks that are 'runny'. The range of storage moduli that were considered printable from the literature are in the range of 10 kPa to 100 kPa, while our experiments showed that the minimum storage modulus required to be printable was 1000 kPa. Since such a large range of storage moduli may be considered printable, it is not a metric that is sensitive to printability across different materials.

Fig. 4(c) shows $\tan(\delta)$, which is equal to the ratio of the loss to the storage modulus. If $\tan(\delta) > 1$ means the ink has a loss modulus greater than the storage modulus, and the material is often said to be more fluid-like. If $\tan(\delta) < 1$, the storage modulus is greater, and the material is often said to be more solid-like. While there are issues with this simplified description of solid-like and liquid-like [45,46], Fig. 4(c) shows that $\tan(\delta)$ is not a useful metric to separate any level of printability.

Finally, Fig. 4(d) shows the axial stress measured by the rheometer. Axial stress here is the normal stress in the direction perpendicular to the parallel plates. For highly-filled inks, during the loading process the axial force will increase while the top geometry is being lowered to the desired geometry gap. Once the desired gap height is reached, the axial force begins to decay to a non-zero value that is ink dependent. This equilibrium value of the axial stress is a measure of the confinement of the filler in the binder at a constant gap height at zero or small shear stresses induced during the loading process. This axial stress remains constant during the application of small-amplitude oscillatory shear, and Fig. 4(d) shows that the axial stress in the linear regime can differentiate printability for all three responses. For inks that are 'runny', the axial stress ranges between 10 Pa and $2.5 * 10^3$Pa. At the lowest end of axial stresses, we see inks that have loadings so small that the loss modulus is larger than the storage modulus in the linear regime. Inks that are 'printable' range from $4 * 10^3$ to $1.5 * 10^4$ Pa, and inks that are 'jammed' have axial stresses above $2.5 * 10^4$ Pa. The magnitude of this axial stress under small shear stress is a good predictor of whether the ink will be printable or not. Since this axial stress is measured after lowering the gap from 1.05 mm to 1.00 mm, we could also define these as an axial or compressive modulus, however, the results will be consistent, as the trim gap and final gap are equivalent among all inks studied. All data used to construct the box plots in Fig. 4, as well as Fig. 6 and Fig. 8, are shown in a Table SM 1.

3.4. Determining a printability criterion at the onset of non-linearity

We have performed a detailed analysis of the linear viscoelastic response of highly-filled pastes, where credible measurements can be utilized to predict printability. As discussed in the introduction, the onset of non-linearity as a function of amplitude is often used to define a yield stress, which is then used to predict printability. One way of determining the yield stress requires finding the stress where the storage and loss modulus cross. However, there are 'printable' and 'jammed' ink formulations that never display a crossover, thus no yield stress could be defined in this way. This lack of a crossover is due to the fact that material leaves the gap before any crossover can occur rather than true rheological properties of the material. Other methods of defining a yield stress involve determining when the storage modulus begins to deviate from linearity, or interpolations of power law fittings. We refrain from choosing a method of determining a yield stress. Here instead, we define a critical stress as the smallest stress amplitude where the strain amplitude changes by a factor of 10 when comparing the strain amplitudes at the beginning and end,

$$\sigma_c = \left\{ \sigma_0 \middle| \frac{\gamma_{0,f}}{\gamma_{0,i}} > 10 \right\} \tag{2}$$

Since the critical stress is defined as the stress amplitude where the strain changes by at least ten-fold, we must be careful to define the critical strain. For this work, the critical strain is the strain amplitude at the beginning of the application of the critical stress:

$$\gamma_c = \{\gamma_{0,i} \quad | \sigma_0 = \sigma_c\} \tag{3}$$

For some inks that were 'runny', the ratio of final strain to initial strain amplitudes was never greater than 10, and in those cases, the critical stress was chosen as the stress amplitude under which the greatest change in strain amplitude occurred. We note that ratio of final to initial strain amplitude for a given stress amplitude may need to be adjusted if smaller step-sizes are taken in the stress amplitude, but a similar trend will occur, where a larger than expected strain response results from the application of the critical stress amplitude. These applied stress and measured strain amplitudes for three inks are shown in Fig. 5. The stress amplitudes correspond to the right axis, and the strain amplitudes to the left axis. The vertical lines separate the figure into three sections. The left section shows the strain amplitude when the applied stress is smaller than the critical stress. The middle section shows the critical stress where for the first time the strain amplitude changes by more than a factor of 10. The right section shows stresses larger than the critical stress. The data for the remaining inks are shown in Figures SM 9 – SM 13.

These definitions require transient measurement of the strain

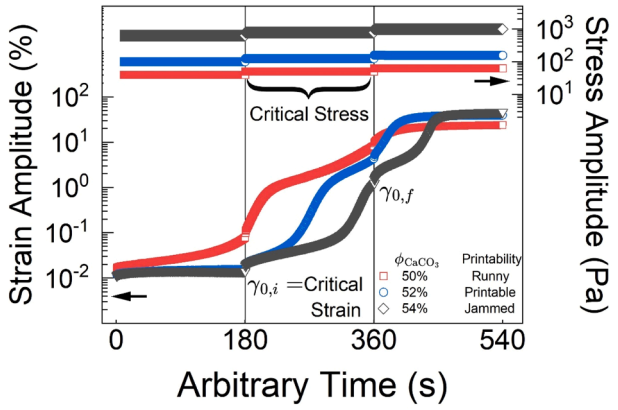


Fig. 5. Applied stress amplitude (right axis) and measured strain amplitude (left axis) of three calcium carbonate inks at 50 vol%, 52 vol% and 54 vol%, that are 'runny', 'printable' and 'jammed' respectively. Three regions separated by vertical lines: $\sigma_0 < \sigma_c$ when the arbitrary time $t_{arb} < 180s$, $\sigma_0 = \sigma_c$ when $180s < t_{arb} < 360s$, and $\sigma_0 > \sigma_c$ when $t_{arb} > 360s$. Critical stress defined as the stress amplitude where $\gamma_{0,f}/\gamma_{0,i} > 10$. Critical strain defined as $\gamma_{0,i}$ when $\sigma_0 = \sigma_c$.

amplitude over time, rather than averaging over multiple periods, like a traditional amplitude sweep in common rheological software. This allows for avoiding ambiguity in determining when the modulus begins to deviate from linearity, or when crossovers occur, and results in a clearer definition. The critical stress may be misrepresented if the stress amplitudes are not applied for enough time. In our experiments, 180 periods at 1 Hz were long enough to determine if a given stress amplitude would cause a change in strain amplitude of at least an order of magnitude.

In Section 3.2, box plots were shown for various linear viscoelastic measures, with the responses separated by printability. Meanwhile, Fig. 6 shows box plots for the critical stress and strain amplitudes for the 'runny', 'printable', and 'jammed' formulations. This well-defined critical stress differentiates between the three printability responses. Inks that have critical stresses < 50 Pa are 'runny', while inks with critical stresses between 60 and 200 Pa are 'printable', and inks with critical stresses > 500 Pa are 'jammed'. As discussed in the introduction, across many works the reported yield stresses of printable materials range from 200 Pa to 2000 Pa, without any distinction of the possible upper limits. This framework gives a range of stresses above which the materials will

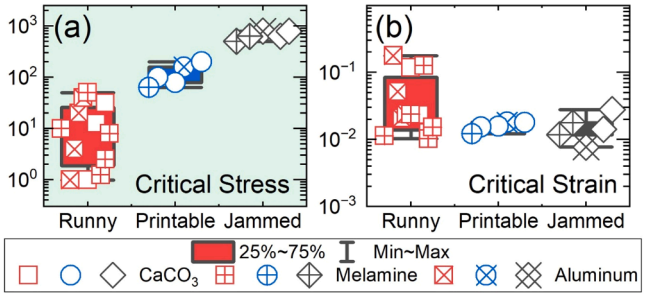


Fig. 6. Box plots showing the applied stress amplitude and measured strain amplitude of each ink immediately prior to non-linearity. (a) is the critical stress before non-linearity occurs (σ_c), and (b) is the critical strain amplitude before non-linearity occurs (γ_c). The critical stress (shaded in green) represents a metric that was able to differentiate between all three ink types. Each filler type is differentiated by the symbol fill, 'X' represents aluminum filler, 'open' represents calcium carbonate, and '+' represents melamine. Eaching printability response is differentiated by symbol shape and color, 'runny' inks are shown by red squares, 'printable' inks by blue circles, and 'jammed' inks by black diamonds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

hold their shape, but below which they can still be extruded.

Interestingly, critical strain amplitude is similar in all the inks measured, around 0.01 %. The few inks that had a larger critical strain are at particle loadings small enough where the loss modulus was greater than the storage modulus across all applied stress amplitudes. For inks that are loaded enough to have some structure at rest, the similar magnitude of critical strains implies that dense pastes are only capable of sustaining a small amount of strain on the rheometer prior to characterizations becoming difficult due to material leaving the measurement gap. This helps to explain the difficulty in utilizing strain-controlled oscillatory procedures or steady-shear experiments to characterize highly-filled DIW inks. Once a small strain is applied, the system undergoes rapid changes as the buildup of internal stresses causes the material to leave the rheometer.

During the application of the critical stress, while the strain amplitude begins to increase, we see the subsequent drop in the axial stress. While this may seem to be related to the yielding of an ink, this cannot be said to be a yield stress or a measurement of the material yielding. The critical stress for highly-filled DIW inks is a measure of the stress above which the rheometer cannot accurately measure the material response because material is leaving the gap. Once the material leaves the gap, all rheological characterizations are not well defined. This is exemplified in Fig. 2 and Fig. 3. We believe that the drop in axial stress is related to cage dynamics for frictional systems [23,60,61]. At rest, each filler particle is surrounded by a cage of other particles. Under the applications of small stresses or strains, particles try to rearrange, but are confined by maintaining a constant gap height. When utilizing two parallel plates, the only option available for the material to relieve these stresses is to leave the gap.

3.5. Determining a printability criterion in relation to maximum particle loading

The maximum packing fraction was determined for each of the particle fillers. Typically, a steady-shear viscosity as a function of volume fraction is fit to an equation, such as the Krieger-Dougherty (KD) relationship [62], to determine the theoretical maximum packing fraction. Since a steady-shear viscosity is unattainable for these pastes, the next choice may be to use the complex viscosity. However, the complex viscosity is calculated using both the storage *and* loss modulus:

$$\eta^* = \frac{\sqrt{(G')^2 + (G'')^2}}{\omega} \tag{4}$$

where G' is the storage modulus, G'' is the loss modulus, and ω is the angular frequency of the applied oscillation, and thus can overestimate the flow behavior when elasticity is dominant [63]. For all DIW inks that are 'printable' or 'jammed' and even some that are 'runny', the storage modulus is greater than the loss modulus, and therefore the complex viscosity would be dominated by the storage modulus, a term that represents the energy stored by a material. Thus, we looked at the loss modulus, more specifically a reduced loss modulus:

$$G''_r = \frac{G''_{SAOS}}{G''_0} \tag{5}$$

where $G''_r$is the reduced loss modulus, G''_{SAOS} is the experimental measurement, and G''_0 is the loss modulus of the resin base. For the exemplar resin used in this study, Sylgard 182, the kinematic viscosity is 5639 cSt at 25°C [49], or $\eta_0 = 5.475$ Pa-s. Then loss modulus would be $G''_0 = \eta_0\omega = 34.4$ Pa. Instead of choosing one of the many possible models to fit the data, the methods of Tanner and Pal to estimate the maximum packing fraction, using the loss modulus by plotting the reduced loss modulus to the power of −0.4.[61,64]. Fig. 7 shows $G''^{-0.4}_r$ vs ϕ, which gives an inverse relationship. The x-axis intercept of this plot is then the maximum packing fraction. The maximum packing fractions are also

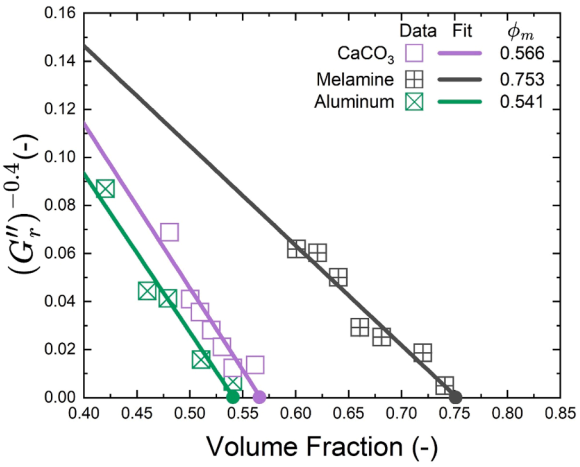


Fig. 7. Linear regime reduced loss modulus raised to the negative two-fifths power plotted against volume fraction for each of the three types of filler, with lines showing the linear relationship for each. The intersection of the fitting line with the x-axis is the maximum packing fraction, indicated by a large, filled circle. The maximum packing fractions for each particle are listed in the legend.

shown in Fig. 7 for each particle. Since the particles have different size distributions, morphology, and surface roughness, their suspensions have different maximum packing fractions.

The maximum packing fraction provides another way to investigate printability, by normalizing the formulation volume fraction by the theoretical maximum volume fraction, as shown in Fig. 8. The normalized packing fraction predicts whether an ink will be 'runny', 'printable', or 'jammed', for the three types of filler with very different morphologies. This leads to a range of normalized packing fractions that an ink is 'printable' of 89.9 % $< \phi/\phi_{\max} <$ 94.3 %. Even if particles have different characteristics, such as distribution and morphology, the normalized packing fraction can predict printability. This trend holds if one chooses to use the linear regime complex viscosity, instead of the linear regime loss modulus. Results using linear regime complex viscosity are shown in Figure SM14 and SM15.

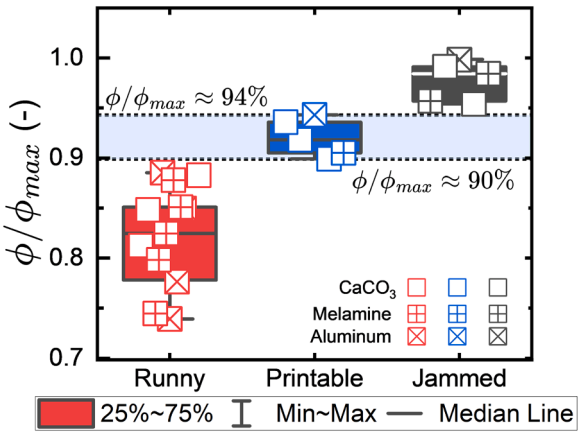


Fig. 8. Box plot showing the normalized packing fraction for all three printability responses. The blue shaded region represents a particle agnostic region of normalized packing fraction where the DIW inks are 'printable', between 90 % and 94 % of the maximum packing fraction determined via the linear fitting shown in Fig. 7. Each filler type is differentiated by the symbol fill, 'open' represents calcium carbonate, '+' represents melamine, and 'X' represents aluminum filler. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Conclusion

We have shown why traditional characterization techniques performed and analyzed by common rheological software have built-in assumptions that yield inaccurate conclusions for dense, highly-filled pastes used in DIW. Specifically, tests that report single values for transient experiments can obscure the true ink response. For steady-shear flow sweeps, researchers apply a range of shear rates to inks, and measure the resulting stress. However, the rheometer does not automatically report the stress over time, and instead reports a single value after the software has determined 'steady-state' has been achieved. Additionally, we have shown why these tests can be difficult to run on dense, highly-filled pastes. Once the critical strain or roughly 0.01 % deformation is achieved, the material leaves the gap to escape the internal stresses caused by particle interactions. Even at small shear rates, well below those experienced by the ink during extrusion, the transient data shows that the measured stress is erratic and meaningless. Rheology measurements were paired with real-time images that show the dense paste deforming and eventually fracturing, creating a slip plane, and squeezing out of the gap.

The inability of highly-filled inks to undergo the application of steady-shear explains why no pre-shear protocol could be utilized for this work. Once the material is on the rheometer, it is allowed to relax under zero shear stress, but any applications of strains larger than the critical strain will lead to degradation of the sample. Since shear rate sweeps are often utilized to determine a yield stress, we have shown that a yield stress is not measurable for these types of materials under steady-shear. Thus, steady-shear an unreliable test protocol for highly-filled DIW inks.

Oscillatory shear is well suited for the task of characterizing DIW inks, ranging from low volume fraction to highly-filled. Specifically, small-amplitude oscillatory shear measurements, which probe the equilibrium structure of DIW inks, without imposing stresses or strains that break the network. These small-amplitude oscillatory shear experiments are easy to make on the rheometer and do not require complicated data interpretations or lengthy time scales, allowing for rapid development of new inks without printing each one.

We present transient oscillatory data for many DIW inks having a range of particle loading and particle types suspended in a silicone base. We show that the rheometer-measured metric that can reliably predict printability is the axial stress. Axial stress is a measure of particle confinement at equilibrium and is directly related to the particle loading of the ink. We show that the axial stress can predict if an ink will be 'runny', 'printable', or 'jammed'.

Meanwhile, other measures proposed for printability, such as the storage and loss modulus could only differentiate between inks that are 'printable' and those that are 'jammed', which may be sufficient for some users. However, these metrics were not able to differentiate inks that are 'printable' and those that are 'runny'. This explains the lack of consensus on a storage modulus that can define printability, as there are inks that are 'runny' but have a larger storage modulus than another ink that is 'printable'.

We show that highly-filled DIW inks have a critical stress and strain amplitude above which rheological characterizations are unsuitable. The ease of small-amplitude oscillatory shear is contrasted by the common methods used to define printability using large-amplitude oscillatory shear. Once the applied stress or strain amplitude of oscillation becomes too large, DIW inks begin to leave the measurement gap, and reported material parameters are corrupted due to inhomogeneous flow. We show this through real-time images where the ink leaves the rheometer under the application of a stress-controlled oscillation. We rigorously define the critical stress through time-resolved oscillatory rheology, as the stress amplitude where the strain response changes by a factor of 10. This definition requires measurement of the unaveraged strain response every period, rather than the average response reported when the rheometer deems the response is at steady alternance. This

critical stress can predict if an ink will be 'runny', 'printable', or 'jammed'. We believe this is different than a yield stress since the ink is not necessarily entering a flowing state but is rather leaving the gap to overcome the frictional stresses induced by particle collisions. Since the ink leaves the gap before flowing, we are unable to characterize the large-amplitude oscillatory behavior of these inks.

Finally, we investigated the relationship between maximum packing fraction and printability. Since steady-shear on dense pastes is impossible, we use the reduced loss modulus in the linear viscoelasic region to quantify the maximum packing fraction for the three fillers. By normalizing each ink formulation's particle loading by the maximum packing fraction for that particle, the printability of each ink is clearly distinguished: inks that are 'runny' ($\phi/\phi_{max} < 89.9\,\%$), inks that are 'printable' ($89.9\,\% < \phi/\phi_{max} < 94.3\,\%$), and inks that are 'jammed' ($\phi/\phi_{max} > 94.3\,\%$). Thus, to predict printability via maximum packing fraction, users need only to make small-amplitude oscillatory shear measurements for a small subset of formulations. From this, the user can determine the theoretical maximum packing for a given particle, and the predicted range of volume fractions that will be printable based on this maximum.

Conversely, this work shows a framework to increase particle loading without sacrificing printability. If the particle filler can be modified to increase the maximum packing fraction, then the volume fraction of the filler can be increased while maintaining the particle packing ratio. Work in the literature has discussed multiple ways to adjust maximum packing fraction [65] including increasing particle size distribution using bimodal or trimodal particle size fractions [66,67]. Recent work also indicates that some log normal particle size distributions result in more efficient particle packing and higher maximum packing fractions [68]. Particle aspect ratio [69,70] roughness [71,72] and morphology [73] are also known to play a role. Dispersants can also play a role in decreasing aggregation and interparticle friction, which increases the maximum packing fraction [74].

Since printability is directly tied to particle loading, for applications where particle loading is the most important quality, changes to filler characteristics prior to ink formulation could increase packing fraction without affecting printability. When it comes to applications, this could lead to higher energy density for batteries, stronger materials for dental composites and concretes, or reducing the need for post-processing densification to remove binder from an ink.

The printability criterion and methodology within may not be suitable for materials such as attractive colloids, where the structure is induced not by frictional jamming, but by surface chemistry. These materials would gel far below their maximum packing fraction and would present issues for this methodology. Additionally, the fillers utilized in this work are all incompressible. If the particles were deformable, flexible, or had large aspect ratios, the metrics within may not be suitable to determine printability for these types of inks.

This work is an important contribution and provides the field of DIW with a method of determining printability of highly-filled inks through a series of small-amplitude oscillatory shear experiments and predicting the maximum packing fraction. By measuring the axial stress induced on the rheometer by the material, or determining the maximum packing fraction, the printability was determined for DIW inks spanning from those that are too thin to hold their shape, to those that are too thick to be extruded, across fillers with different shapes and sizes. Additionally, this work has opened the door to do quick printability screenings of various formulations across particle type, morphology, and size distribution.

CRediT authorship contribution statement

Simon A. Rogers: Supervision, Project administration. **Alexander S. Tappan:** Project administration, Funding acquisition, Conceptualization. **Jessica W. Kopatz:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **James John Griebler:** Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation. **Anne M. Grillet:** Writing – review & editing, Writing – original draft, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.addma.2024.104578.

Data availability

The authors do not have permission to share data.

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